Problem 1 : $P(R \text{ is good}) = 0.6 \quad P(R \text{ is bad}) = 0.4$

$P(\text{long line} \mid R \text{ is good}) = 0.9$

$P(\text{long line} \mid R \text{ is bad}) = 0.2$

$$P(R \text{ is good} \mid \text{long line}) = \frac{P(\text{long line} \mid R \text{ is good})\, P(R \text{ is good})}{P(\text{long time})}$$

$$= \frac{P(\text{long line} \mid g)\, P(g)}{P(\text{long line} \mid g)\, P(g) + P(\text{long time} \mid b)\, P(b)}$$

$$= \frac{0.9 \times 0.6}{0.9 \times 0.6 + 0.2 \times 0.4} = 0.87$$

Problem 2 : $\Omega = \{1, 2, 3, 4\}^2$

$$P(A) = P\{\min(X, Y) = 2\} = P\{(2,2), (2,3), (2,4), (3,2), (4,2)\} = \frac{5}{16}$$

$$P(B) = P\{\max(X, Y) = 2\} = P\{(1,2), (2,2), (2,1)\} = \frac{3}{16}$$

$$P(A \cap B) = P\{(2,2)\} = \frac{1}{16}$$

$\therefore P(A) \cdot P(B) = \frac{15}{16^2} \neq P(A \cup B)$

$\therefore$ Event A and B are not independent.